Sales and Inventory Analytics for Precast RCC Products

A Final report for the BDM capstone Project

Submitted by

Name: Krishna Goel

Roll number: 23F3003674

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

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1) Executive Summary

This project focuses on analysing two critical business challenges faced by the business, **Laxmi Concrete Products** that are Moderately Unpredictable Demand and Payment Delays and Default Risk. The main objective of this study is to use business data to support better inventory planning and improve cash flow.

Historical sales and Inventory data for 10 SKUs from September 2024 to December 2025 were analysed to understand the demand behaviour. Demand forecasting on monthly basis was performed using MS Excel Forecast Sheet tool. The results reveal that demand significantly varies across SKUs, where some SKUs show relatively stable demand compared to other SKUs that show sharp fluctuations. An ABC Analysis was also conducted to improve inventory planning and decision making. The analysis reveals that Category A SKUs contribute nearly 80% of total revenue, this shows that business depends heavily on small number of SKUs, while Category C SKUs contribute minimally.

In addition to sales and inventory analysis, customer payment behaviour was analysed using debtor data for 29 clients for financial year 2025-2026. Key performance indicators like Days Sales Outstanding, Outstanding Amount Proportion, Collection Efficiency were calculated to assess payment default risk. The results indicate that nearly two third of the clients fall under High Risk category, characterised by long payment delays, high outstanding amounts and old dues. This indicates delays in cash inflows and an increased risk of payment defaults.

Overall, the study shows that although the business has strong revenue generating SKUs, but operational efficiency is constrained by demand uncertainty and weak receivables management. Based on findings, it is recommended that business should adopt selective inventory control by prioritizing high value SKUs and use demand forecasting for procurement planning. Moreover, business should implement risk-based credit policies, stricter payment terms for high-risk clients and focused recovery of old dues to improve cash flow and reduce the payment default risk.

2) Detailed Explanation of Analysis Process

2.1) Analysis Process for Moderately Unpredictable Demand

Data Collection and Preprocessing

To tackle with unpredictable demand Historical sales and inventory data was collected for 10 SKUs from the business ledgers and organised into the MS Excel sheet. Each SKU was structured in a separate sheet with following format i.e.

- Date
- Opening Quantity
- Purchase Set
- Sale Quantity
- Sale value
- Closing Quantity

Initially six-month data was collected from September 2024 to February 2025, then further data up to December 2025 was collected to improve the analysis and find better insights. The daily sales showed irregular pattern so the data was preprocessed by aggregating it into month wise. This month wise data helped to identify the overall monthly demand and reduces daily variability.

Demand Forecasting

After aggregating the SKU sales data month wise, the demand of each SKU was forecasted using the excel forecast sheet tool. The Forecast sheet tool is present inside the Data tab under the forecast section and this tool uses Exponential smoothing, which is time series forecasting method that gives more importance to recent data while still considers past values. Exponential Smoothing assigns decreasing weights to older values, so recent sales trend have a stronger influence on predicted sales. This method combines previous forecast values with actual observed values and adjusts forecast based on recent errors, allowing the model to quickly adapt to changes in demand patterns making it more suitable for this data where demand is not stable. The Forecast Sheet tool automatically detects seasonality in data and interpolates any missing data. It also generates confidence intervals which provide an upper and lower range of expected demand. This tool is also computationally lighter than Arima or Sarima models and doesn't require to manually identify p, q parameters for predicting future demand. The tool requires Value series that is historical demand of the SKU and a Timeline series i.e. monthly intervals of sales data like September 2024, October 2024 up to December 2025. Then monthly sales from September 2024 to December 2025 were arranged in chronological order for every SKU then a line chart was created to visualise the historical demand of the SKUs. Then the excel forecast sheet tool was applied onto the sales data from September 2024 to December 2025. This tool automatically identified the seasonality of data and confidence level of 95% was selected to predict the demand. The tool then forecasted monthly demand value for each SKU for months January

to April 2026 with upper and lower confidence and it also created a line chart for the visualization of sales from September 2024 to April 2026. This method was employed as it simple, and practical for small business decision making. The forecasted demand would help business to maintain an optimal inventory as per demand that would avoid situations of overstocking and understocking.

ABC Analysis of SKU

After forecasting monthly demand for each SKU, the ABC analysis was performed on to the data to identify the relative importance of each SKU. For ABC analysis total sale quantity and total sale value for each SKU was collected for given period and both of these was considered while analysis. Then the SKU were arranged in descending order i.e. the highest sale valued SKU was ranked first and lowest sale valued SKU at the last. Then cumulative percentage contribution of each SKU was calculated. Based on the calculated contribution the SKU were classified into 3 categories -

- **Category A**: High demand item that contributes to majority of sales. They contributed almost 80% of the total revenue. Example Manhole cover with frame HD-20 and Manhole cover with frame EHD-35 contributed to almost 80% of the total revenue.
- **Category B**: Moderate demand item that contributes up to 15% of the total revenue. Example Bollards, Manhole cover with frame MD-10 and RCC Slabs that contributed 15% of the total revenue.
- **Category C**: Least demand items with minimal contribution of less than 5% to sales and revenue. Example Bench, RCC pipes contributed least to the total revenue.

SKU that belongs to category A require strict monitoring and regular replenishment. SKU belonging to category B require periodic review and SKU belonging to category C SKU should be stocked cautiously so that they don't overstock and increase the inventory cost of the business. This classification will help to perform selective inventory control where business can focus more on high value items rather than low value item that doesn't contribute enough to sales.

2.2) Analysis Process for Payment delay and default risk

Data Collection and Preprocessing

To analyse the customer payment behaviour the debtor's data for financial year 2025-2026 was collected for 29 different clients from the business records. Data ranges from April 2025 to February 2026 representing total of 335 days. The collected data was initially organised month wise in following format for each client i.e.

- Opening Balance
- Debit (credit sales)
- Credit (collection received)

- Closing Balance.

Then this data was consolidated client wise to create summary table consisting of the following fields i.e.

- Client Name
- Year opening Balance
- Total sales
- Total Collection
- Year End closing

Calculation of Receivables Performance Indicators

Average Receivables: This represent the average amount of money that is outstanding for a given period. In this case the average receivables were simply calculated as sum of opening and closing balance by 2. If the opening balance in credit, then it was taken as negative because it was the advance payment that was given by client. if the opening balance was in debit, then it was taken as positive because it is the amount that the business needs to collect.

$$\text{Average Receivables} = \frac{\text{Opening Balance} + \text{Closing Balance}}{2}$$

Days Sales Outstanding: DSO measures the average number of days taken by a customer to pay the dues of business. If no sales are generated and there is outstanding amount then it is treated as old dues and it considered to be a high risk.

$$\text{DSO} = \frac{\text{Average Receivables}}{\text{Total Sales}} \times \text{Days}$$

In case of Laxmi Concrete Products number of days are 335.

Outstanding Amount proportion: Outstanding Amount proportion measures the proportion of Year end closing balance by sum of year opening balance and total sales. It helps to determine the Outstanding amount that need to be recovered from the clients.

$$\text{Outstanding Proportion} = \frac{\text{Closing Balance}}{\text{Opening Balance} + \text{Total Sales}}$$

Collection Efficiency: Collection Efficiency measure the proportion of total dues that were collected during the given period. The collection efficiency can be calculated as Total collection made by the sum of Opening Balance and Total sales.

$$\text{Collection Efficiency} = \frac{\text{Total Collection}}{\text{Opening Balance} + \text{Total Sales}}$$

Payment Default Risk Assessment Based on Payment Behaviour

Outstanding Amount proportion Based Default Risk

The default risk was calculated based on the percentage of outstanding amount proportion for each client. This risk signifies how much percentage amount is unpaid by the client. On the basis of outstanding amount proportion, the clients were divided into 4 categories as follows

- **No Risk** - When there is no outstanding amount.
- **Low Risk** – When less than 12.5% of the amount is outstanding.
- **Medium Risk** – Outstanding Amount is between 12.5 to 25%.
- **High Risk**- Outstanding Amount is more than 25%.

Days sale Outstanding based Default Risk

The risk was calculated based on the Days sales outstanding. This risk is Time based that signifies how long the payments are delayed by the clients. On the bases of delays the clients are categorised as follows

- **No risk** - when days sales outstanding is 0 days.
- **Low Risk** - when DSO is less than 60 days.
- **Medium Risk** - When DSO is between 60 to 120 days.
- **High Risk** – when DSO is more than 120 days.

Final Risk Assessment

Since the client's behaviour could not be judged based upon a single indicator hence, both indicators were used to calculate the payment default risk. The client's risk level was determined by selecting higher risk between the Outstanding amount-based risk and DSO based risk. This approach ensures that client with either long payment delays or with large outstanding amount are classified appropriately. Moreover, this provides a comprehensive view of customer credit behaviour by considering both Amount based risk and Time-based risk.

$$\text{Final Risk} = \max(\text{Outstanding Risk}, \text{DSO Risk})$$

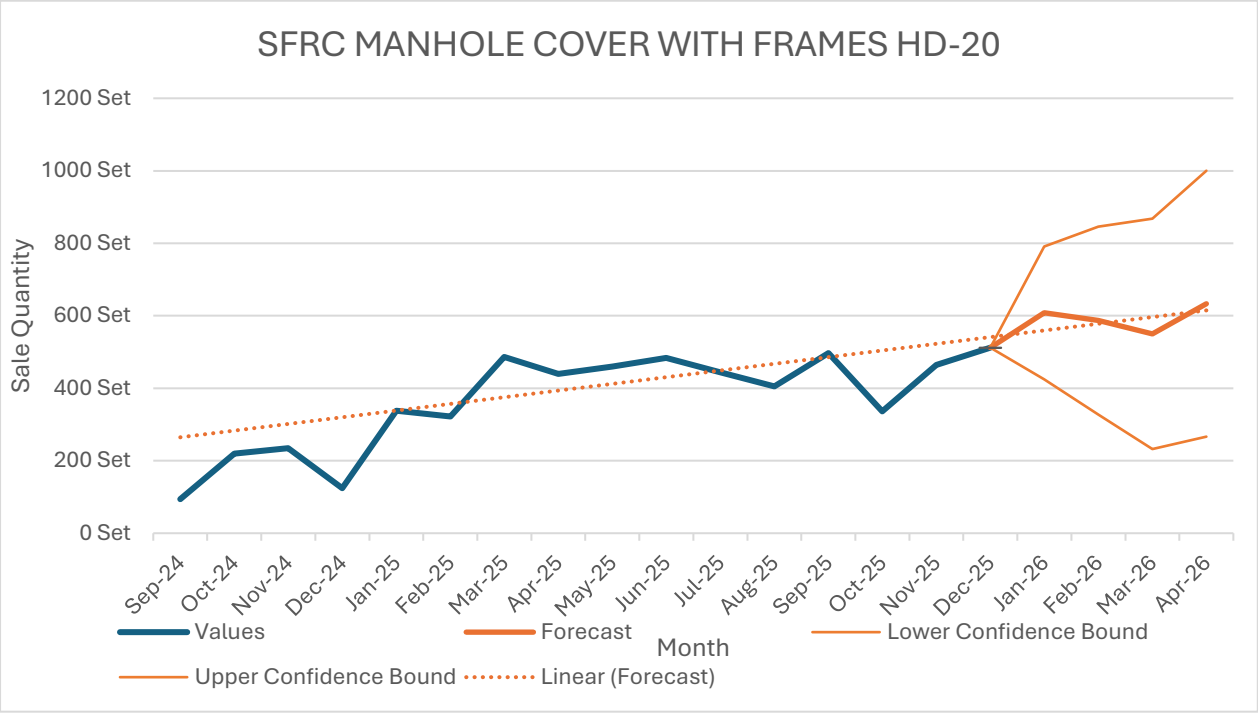
3) Results and Findings

The data collected from Laxmi Concrete Products was analysed using Microsoft Excel to address the two major challenges that are demand uncertainty and payment delay and defaults. These results are based on historical sales data, and customer ledger data. To identify the pattern and trends in the data various charts were utilized. The finding is divided into two parts as per the problem statement. First part focus on the demand behaviour and inventory management, where the second part evaluates clients based on their payment behaviour.

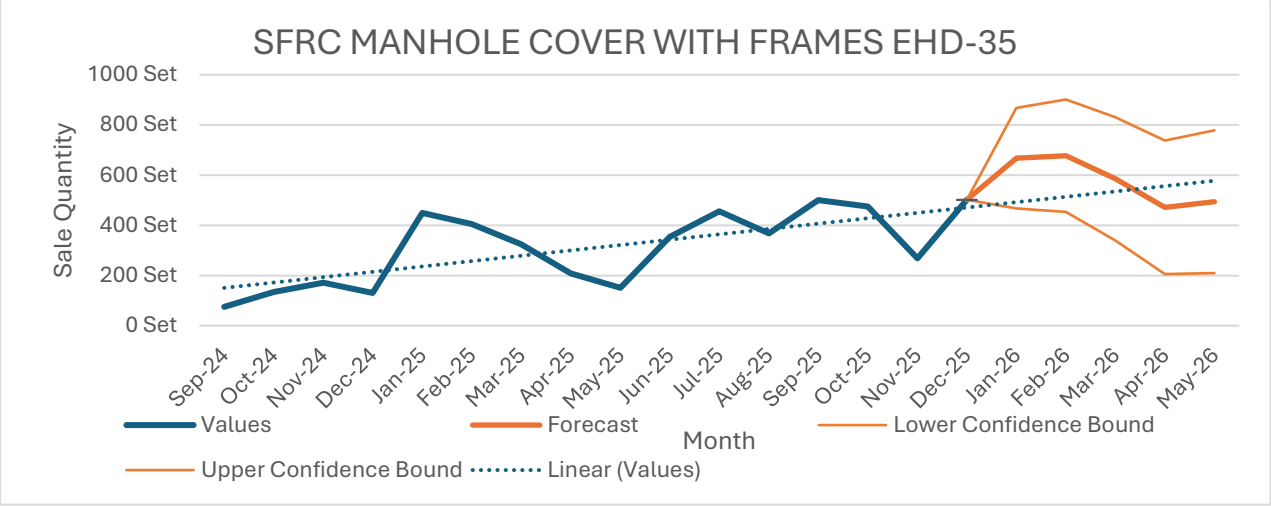
3.1) SKU Wise Demand Forecasting

Demand Forecasting for ten SKU were performed on to the historical sales data ranging from September 2024 to December 2025. This was done to estimate the near future demand for each SKU. The forecasting was done using MS Excel Forecast Sheet Tool with 95% confidence level. The following charts present the Forecasted demand along with the historical demand to better visualize the sale pattern for each SKU. Also, these represent the demand behaviour and highlight the variations in sales along with upper and lower confidence. This analysis shows that the demand for RCC products is not stable and varies across different months.

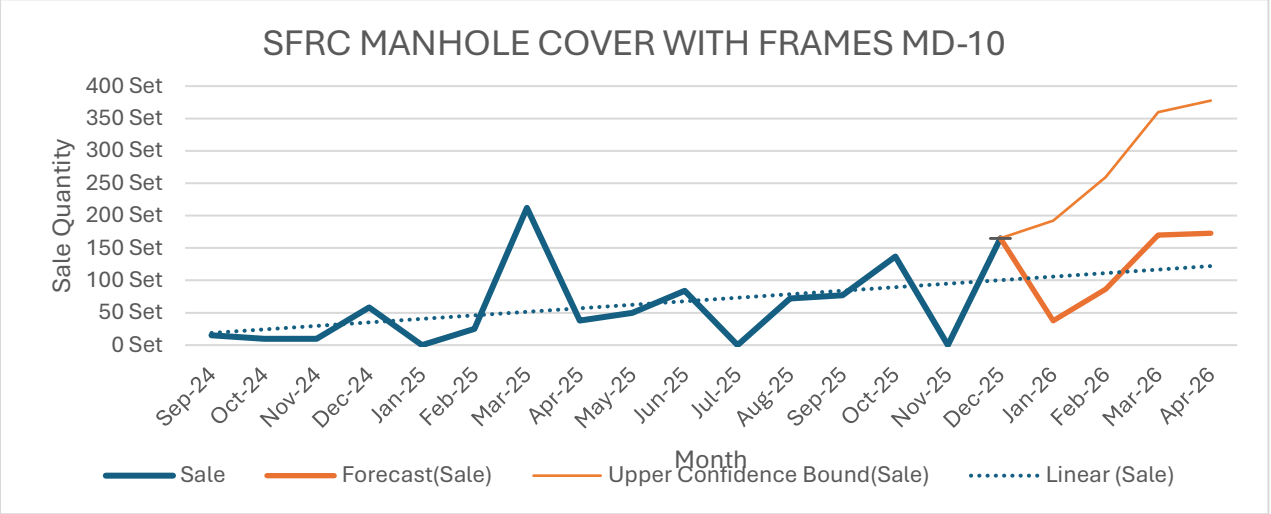
SKU wise sales forecast charts are presented below:



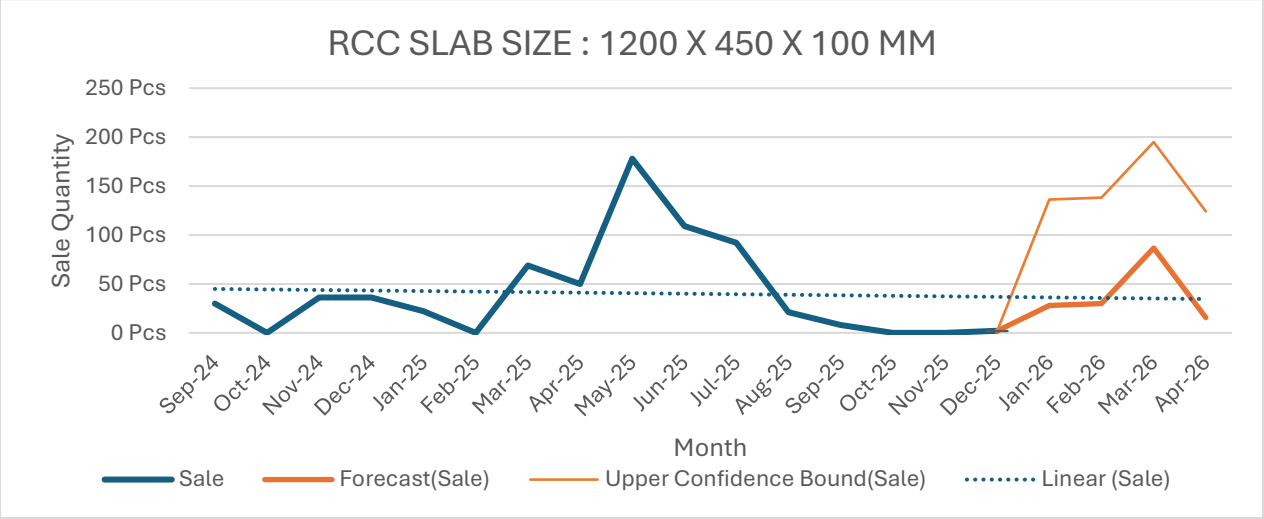
Graph 1 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for SFRC MANHOLE COVER WITH FRAMES HD-20***



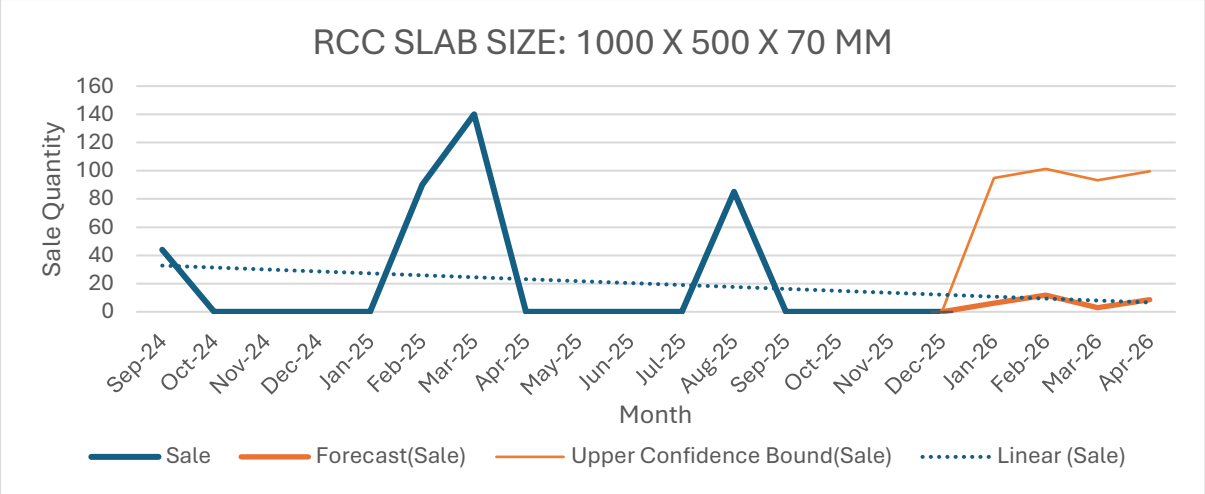
Graph 2 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for SFRC MANHOLE COVER WITH FRAMES EHD-35***



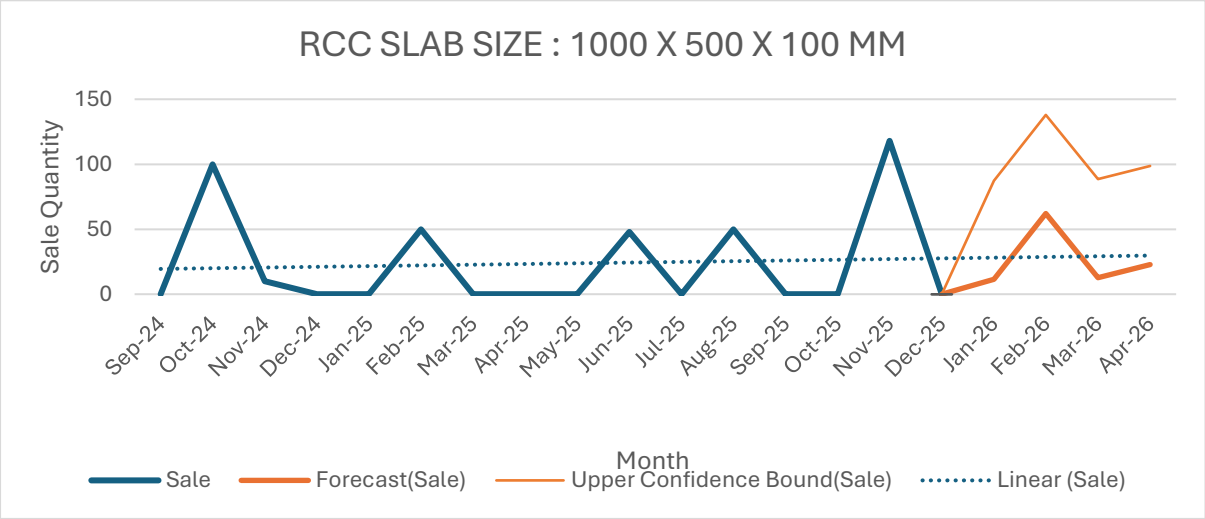
Graph 3 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for SFRC MANHOLE COVER WITH FRAMES MD-10***



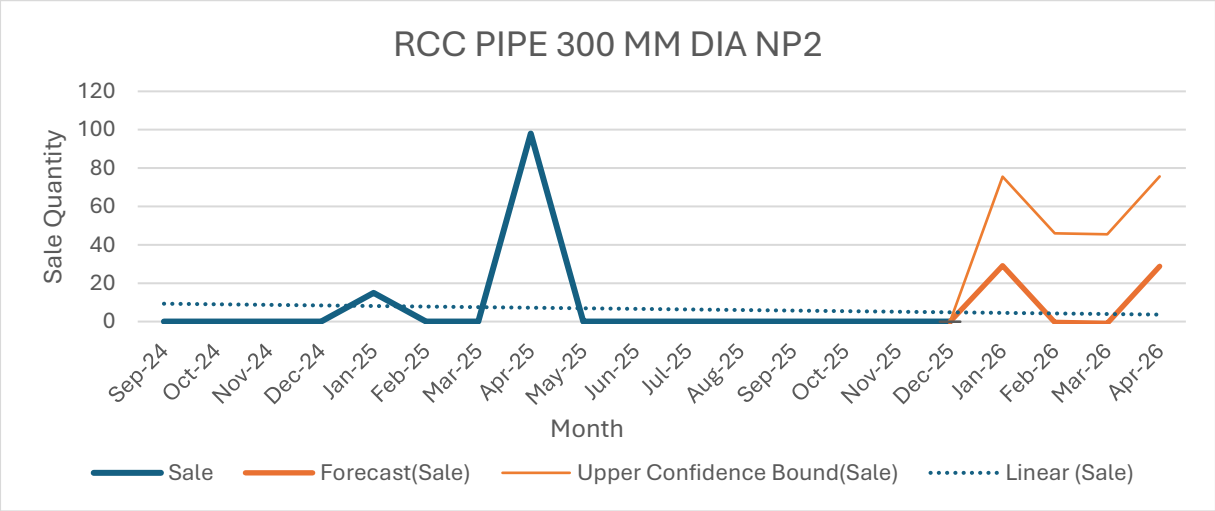
Graph 4 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for RCC SLAB SIZE: 1200 X 450 X 100 MM***



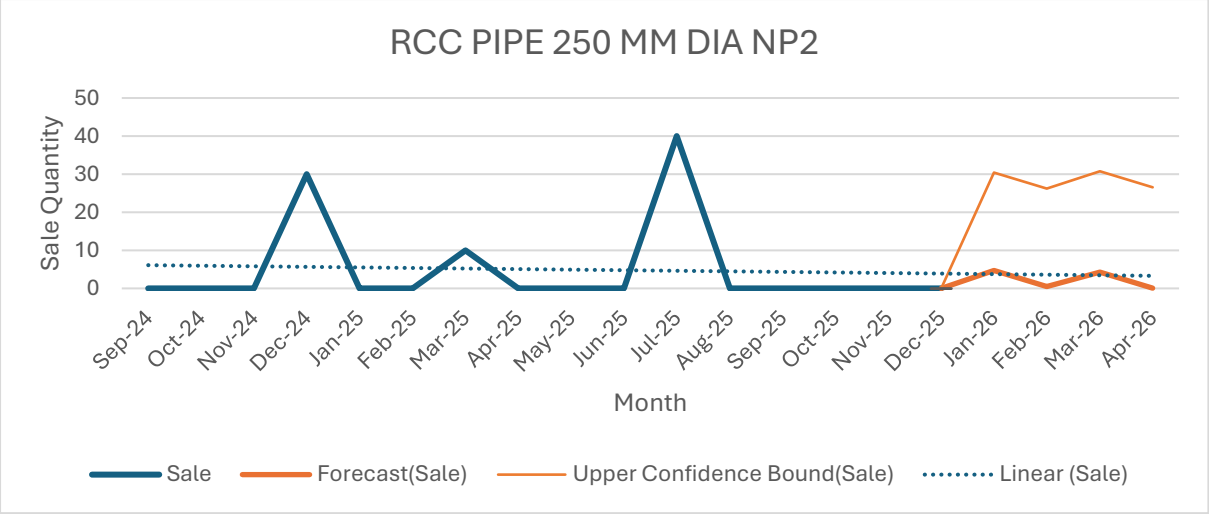
Graph 5 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for RCC SLAB SIZE: 1000 X 500 X 70 MM.***



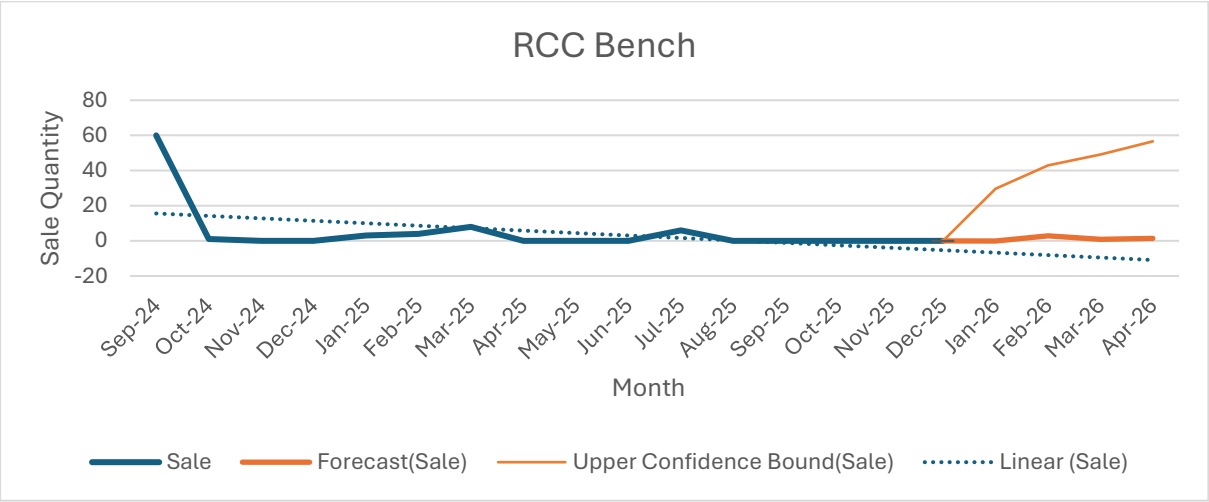
Graph 6 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for RCC SLAB SIZE: 1000 X 500 X 100 MM***



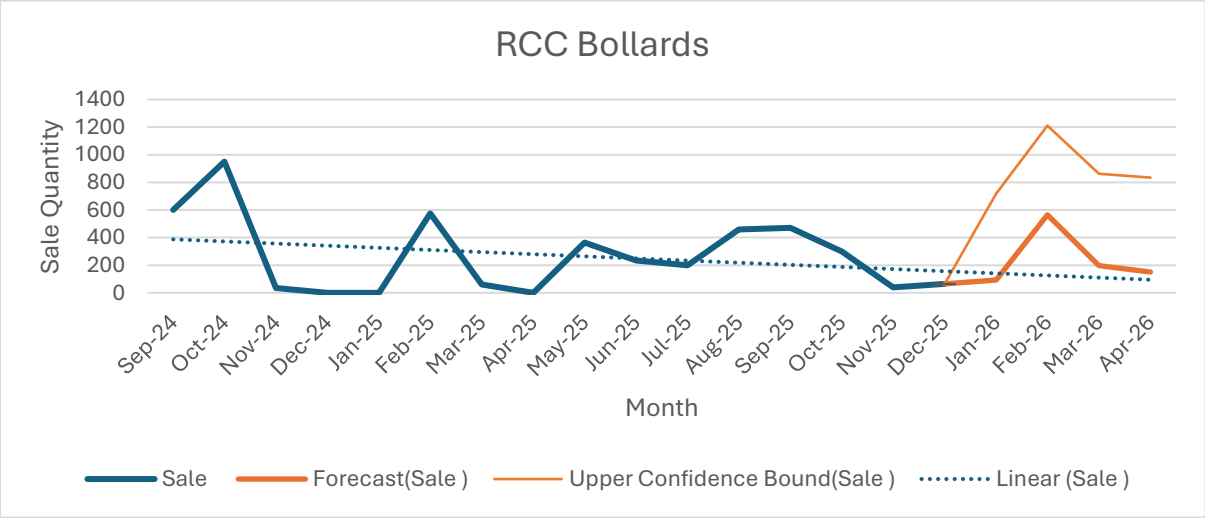
Graph 7 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for RCC PIPE 300 MM DIA NP2***



Graph 8 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for RCC PIPE 250 MM DIA NP2***



Graph 9 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for RCC Bench***



Graph 10 ***Sales Analysis*** *(September 2024 to December 2025)* ***and Forecasting*** *(January 2026 to April 2026)* ***for RCC Bollards***

A detailed view of historical demand along with projected demand can be seen from above forecast charts. This analysis indicates that the demand for RCC products is not uniform and varies significantly across different month and SKUs. This confirms that there is moderately unpredictable demand in Laxmi Concrete Products.

The data represents sharp fluctuations in the monthly demand for several SKUs. SKUs such as RCC Slabs and Pipes shows irregular demand pattern, where some months record very little sales or even zero sales after certain spikes in demand. Such sales reflect that demand is not consistent throughout the year.

SKUs like SFRC Manholes covers with frames (HD-20 and EHD-35) shows higher and relatively consistent demand than other SKUs. The forecast result show that these products will maintain their demand level in the near future. On the other hand, the product like RCC slabs, Pipes and Benches show lower average sales with higher variability in demand. The forecast chart for these products has wider confidence interval i.e. distance between upper and lower bounds, indicating higher uncertainty in the future.

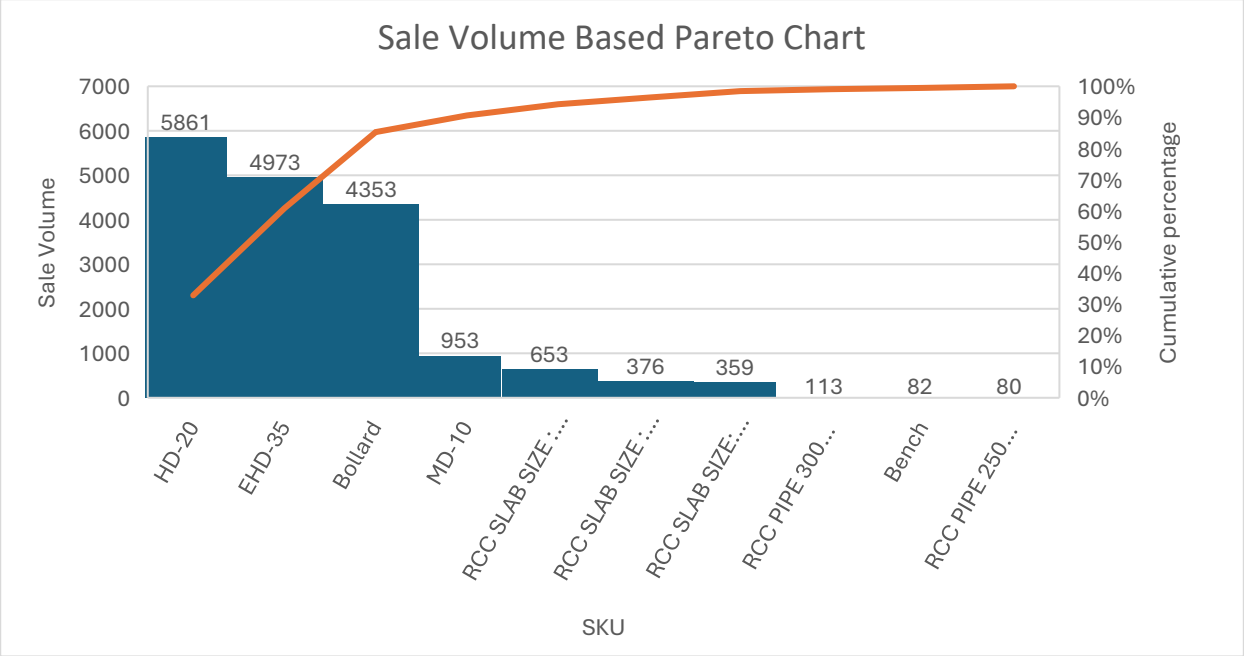
The trend pattern across the SKUs further indicate that demand does not follow uniform pattern and is influenced by external factors like project schedules, government polices related to construction and seasonal construction activity. The presence of irregular sale cycle suggest that average based inventory approach should not be followed for all product as it may cause over stocking of less selling products.

The Upper and lower confidence limits in the sales forecast charts represent the expected range of future demand. For SKUs like RCC Slab and Pipes the confidence is wider, indicating higher uncertainty in demand in near future. Whereas SKUs like Manhole (HD-20 and EHD-35) show narrower confidence band, reflecting more stable demand in near future.

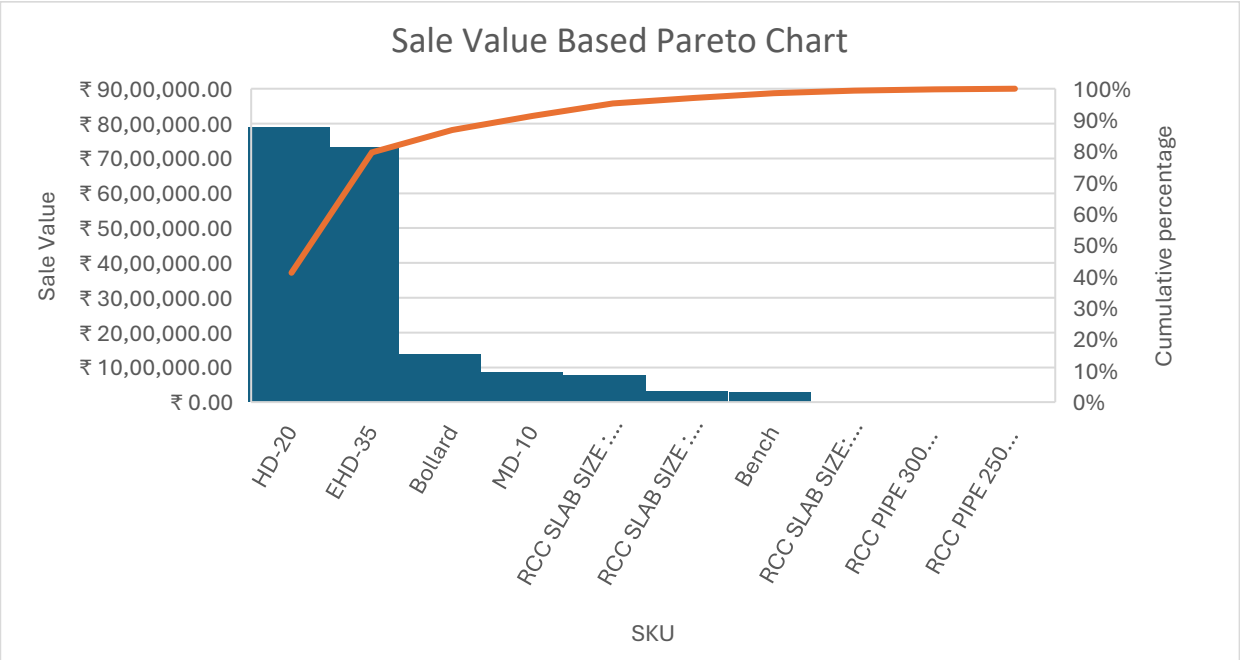
Overall, the graphs indicate that demand variability exists and differs significantly across SKUs. Inventory planning should not be based on uniform stocking approach as it may lead to overstocking of less sold SKUs.

3.2) ABC Analysis

An ABC Analysis was performed on the SKUs based on total sales volume and total sales value to strengthen inventory decision making under Moderately Unpredictable Demand. This analysis helps in enabling selective inventory control instead of uniform stocking for each SKU by identifying critical SKUs that contribute more to sale and revenue. The ABC analysis was carried out using cumulative percentage contribution of each SKU towards revenue. SKUs were arranged in descending order of their revenue contribution and then classified into A, B and C categories. Where Category A represent highest contributors, B category represent moderate contributors and category C represent least contributors to sales and revenue.



Graph 11 Pareto Analysis of sales volume by SKU



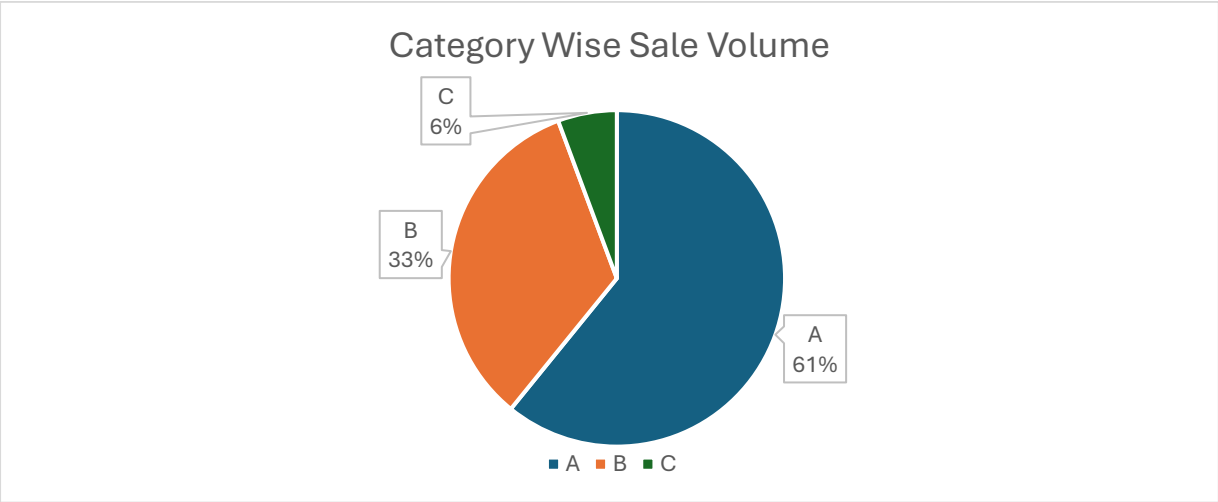
Graph 12 Pareto Analysis of sales value by SKU

The pareto chart result shows that Manhole cover with frame (HD-20 and EHD-35) contribute highest share to both total sales value and sales volume. Together these two SKUs account for large portion of cumulative contribution nearly 80% of total sales value and dominate overall revenue generation for Laxmi Concrete Products. Hence, they fall under category A. Also, the sales volume based pareto chart indicate that these SKUs record highest sales quantities than other SKUs. This indicate that small number of SKUs contribute majorly to both sales and revenue generation for the business.

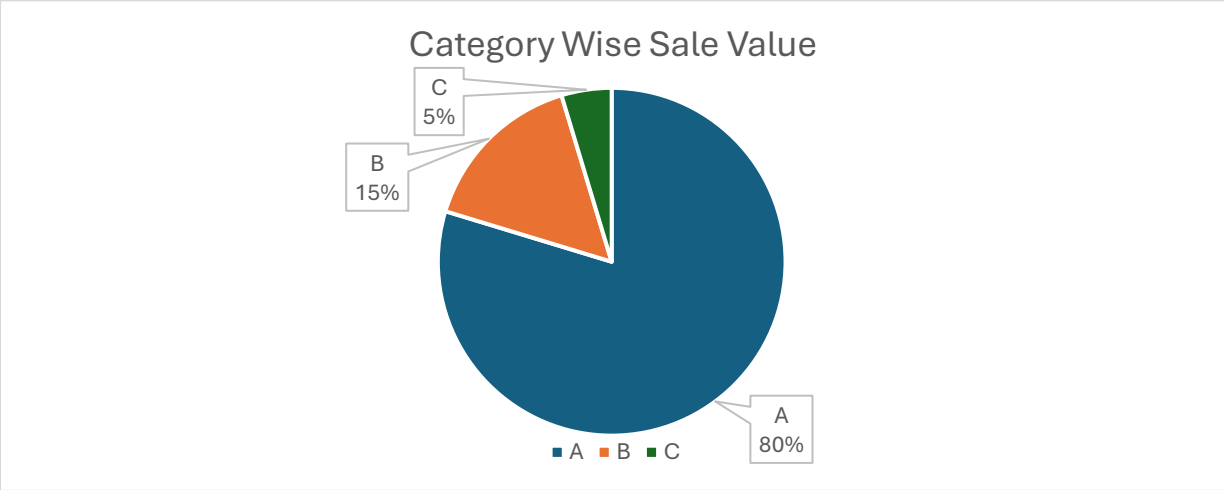
SKUs like Bollards, Manhole cover with frame MD-10 and RCC slabs contribute moderate share to total sales volume and sales value. Altogether these SKUs account for approximately 15% of total cumulative contribution towards revenue. Hence, they are categorised as Category B. The contribution of this category is less than category A but significantly higher than category C.

RCC Benches and RCC pipes contribute least to the total sales volume and sales value, together they account for less than 5% of the total cumulative contribution hence, they can be classified as category C. Pareto chart represents that these SKUs contribute to very low sales and revenue. Moreover, their cumulative percentage increases marginally towards end of the pareto curve.

The pareto distribution is observed in both sales volume and sales value where small number of SKUs (HD-20 and EHD-35) account for majority of sales and revenue while other SKUs contribute very little. This analysis provides a structured classification of SKUs in A, B and C category based on their relative contribution towards sales and revenue for Laxmi Concrete Products.



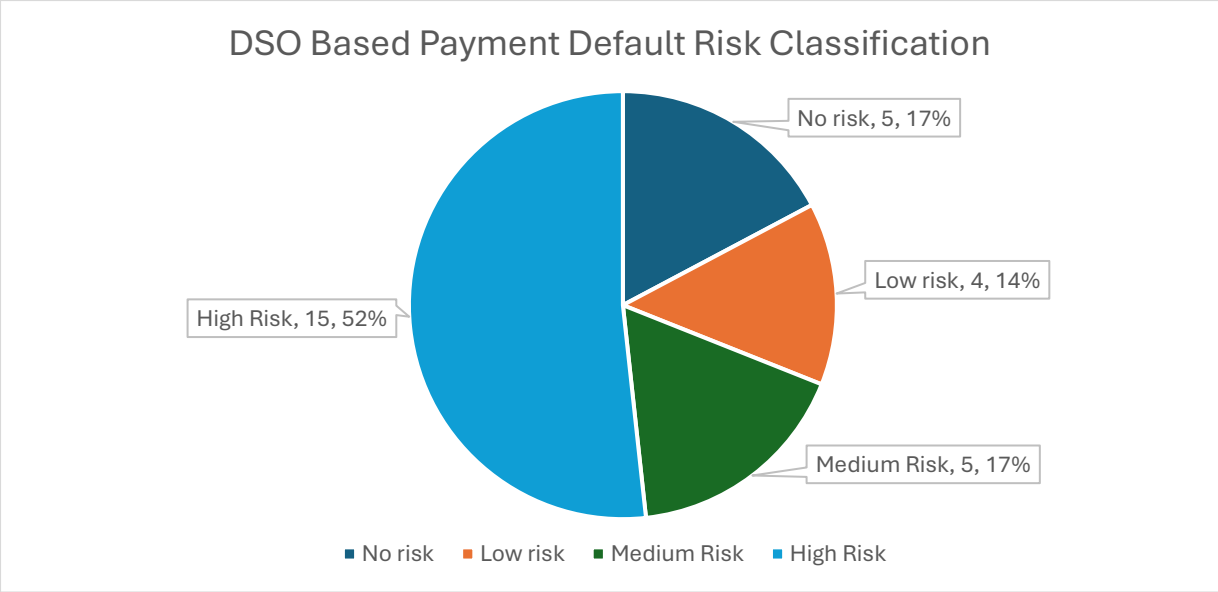
Graph 13 Proportion of total sales volume across Categories



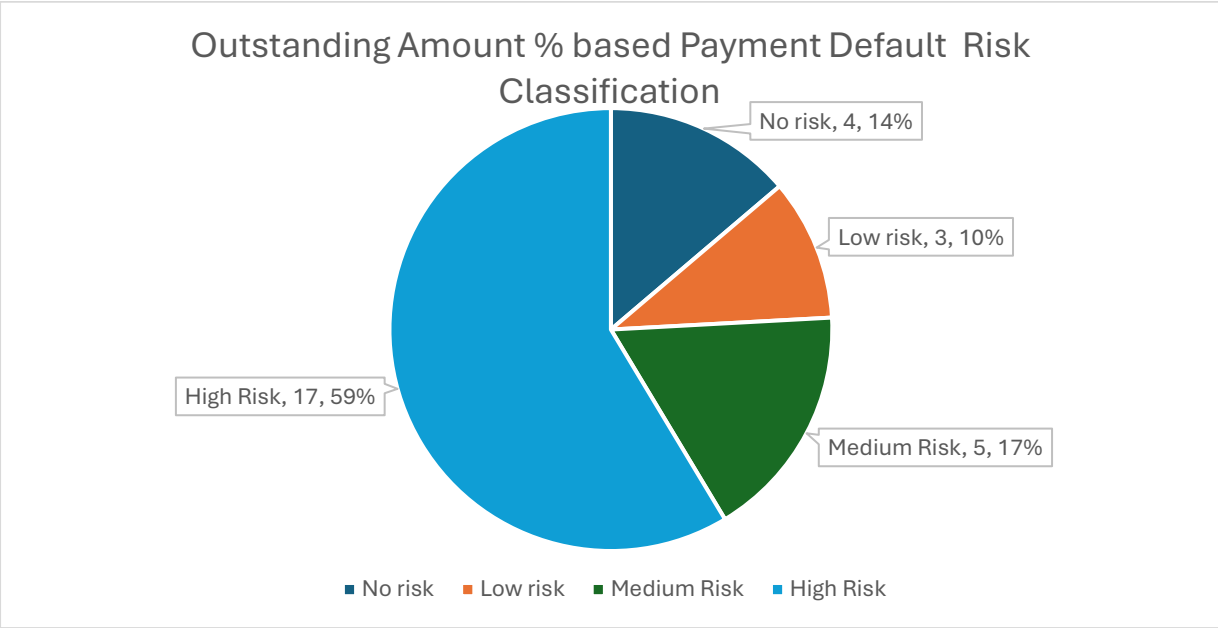
Graph 14 Proportion of total sales value across Categories

3.3) Payment Delay and Default Risk

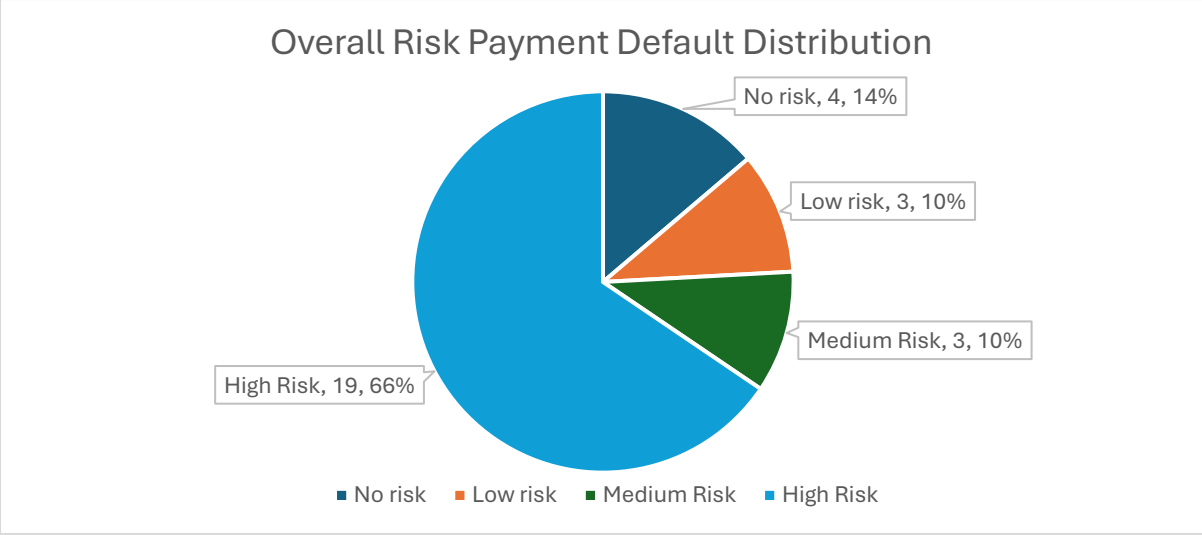
Customer ledger data for 29 clients was analysed to evaluate payment delay and default risk for financial year 2025-26. The analysis was conducted using risk indicators like Average Receivables, Days Sales Outstanding (DSO), Outstanding Amount Percentage, and Collection Efficiency that were derived using Year Opening balance, Total Sales, Total Collection, Year-end closing balance. Based on these indicators the risk level for each client was identified.



Graph 15 Distribution of Payment Default Risks by Days Sales Outstanding



Graph 16 Distribution of Payment Default Risks by Outstanding Amount proportion



Graph 17 Overall classification of payment defaults by risk level

The results indicate that there is large variation in payment behaviour across clients. Some clients make timely payment with zero outstanding balance at year end, whereas other clients show significant delays in payment, high outstanding proportions, no collection activity during the year. Few clients have zero year end closing indicating full recovery of payment while, most of the clients carry some outstanding balance. The cases where no sale occurred throughout year but opening balance remain unpaid are considered to be old dues and classified under High Risk Category.

Days Sales Outstanding value varies among the clients; 5 clients have DSO equal to zero indicating timely payments and these clients are classified under No Risk category. 4 of clients fall under Low Risk (14%) and 5 clients fall in Medium Risk (17%) DSO range (1-120 days), indicating moderate delays in payment. While most of the clients (52%) exhibit high DSO value i.e. more than 120 days reflecting prolonged payment delay or old dues, these clients fall under High Risk category. The DSO based Risk assessment shows that High Risk clients form largest group followed by Medium and Low Risk categories.

Outstanding Amount Percentage indicates the proportion of unpaid balances. 4 clients show zero percent outstanding reflecting complete collection of sales; hence these clients fall under No risk category. 3 clients fall under Low Risk (10%) and 5 clients fall under Medium Risk (17%) category indicating that outstanding amount lies below 25%. Whereas 17 clients show outstanding percentages above 25% and fall under High-Risk Category. This represents that unpaid amounts are concentrated among some specific clients instead of being evenly distributed.

Finally, the clients were classified by combining the risk level from Days Sales Outstanding and Outstanding Amount percentage where the higher level of the two was considered as Final Risk level. The Overall Risk Distribution represent that majority of clients are falling under High Risk followed by No Risk, Low Risk and Medium Risk. The distribution highlights that payment risk is clustered among significant number of clients as reflected in the Overall Risk Distribution.

4) Interpretation of Results and Recommendations

The analysis carried out provide meaningful insights for both Moderately Unpredictable demand and Payment Delay and Default Risk at Laxmi Concrete Products. These results together highlight key areas where management attention and improvements are needed.

4.1.1) Interpretation of Demand Forecasting Results

Demand forecasting results indicate that SKUs demand is uneven and moderately volatile across the analysis period. Where few SKUs exhibits relatively stable sales pattern, other SKUs show sharp fluctuations in the demand indicating project-based demand in construction sector. This variability in demand suggest that demand is not purely trend driven and may be influenced by external factors such as infrastructure project cycles, seasonal construction activity and government policies like GRAP where the construction projects were temporarily suspended during peak winters in Delhi NCR resulting into lowers sales in November and December every year.

The forecast output also represents that uncertainty in demand differs significantly across SKUs. SKUs with higher and relatively consistent sales demonstrate narrower forecast range, while low sale SKUs exhibit higher forecast error. This implies that demand is not uniformly distributed among all the SKUs and they must be manged in a selective manner instead of managing them uniformly. For example, more focus should be on high sale SKUs like Manhole cover with frames (HD-20 and EHD-35) followed by bollards and other SKUs. Least focus should be RCC pipes and bench that provide significantly less contribution to sales and revenue.

4.1.2) Interpretation of ABC Analysis Results

Results of ABC Analysis reveal that there is highly skewed contribution pattern within SKUs. Category A items though are in limited number but contributes dominant share both in sales volume and sales value. Nearly 60% of the total sales and 80% of total revenue is generated by category A SKUs alone. Category B SKUs accounts for moderate share of sales and revenue while, Category C SKUs contributes less than 5% of total sales and revenue to Laxmi Concrete Products. This confirms the pareto principle that nearly 80% of business results come from 20% of the products.

The results indicate that business should maintain high availability of category A SKUs as these SKUs contribute nearly 80% to sales and revenue and their shortage can affect business performance. The category B SKUs should be stocked moderately, whereas Category C SKUs should be stocked cautiously or if possible then ordered on demand as these may result into increased holding cost without significant revenue benefits to Laxmi Concrete Products.

4.1.3) Interpretation of Payment Delay and Default Risk Results

Payment delay and default risk results show that payment behaviour of clients is highly uneven where few clients settle their dues on time while, large number of clients show significant delays that affect company's cash flow.

Out of 29 clients only 4 clients fall under No risk and 3 fall under Low-risk category meaning that they clear their dues within acceptable time and have low outstanding balance. These clients show generally low DSO and less than 12.5% of outstanding amount proportion indicating a disciplined payment behaviour. 3 Clients fall under Medium Risk Category indicating that outstanding amount is in between 12.5-25% or DSO is in range 60-120 days. Whereas, more than half of the clients (19) are classified under High-risk category based on DSO and Outstanding amount criteria, which implies that they haven't cleared dues for more than 120 days or have more than 25% of the amount as outstanding. In some cases, the DSO was more than 365 days indicating prolonged delays in payment. Also, multiple cases show that many of the clients have outstanding amount proportion more than 50% that indicates that large proportion of sales revenue was unrealised in form of receivables, directly affecting the cash flow at Laxmi Concrete Products.

Some clients have Opening balance and no sales activity and yet the outstanding amount remains unpaid classifying them into old dues. These cases suggest long pending receivables and represent blocked working capital with high risk of payment default. Collection efficiency results also highlight weak recovery performance for most of the clients with collection efficiency lying below 50% and some have 0% indicating complete non recovery during the year. This uneven collection pattern represents that recorded revenue does not always turn into cash.

Overall risk assessment that combines both time-based and amount-based risk measures show that almost two third of the clients fall under High risk category. This large concentration of High-risk clients indicate that payment delays may be a systematic issue rather than some exceptional cases. This exposes the firm into slower cash flow and possible defaults. While the sales are generated via multiple clients, a large portion of revenue is stuck in receivables, increasing financial risk. This analysis highlights that the firm should view client's performance based on payment reliability instead of sales volume alone.

4.2) Recommendations

On the basis of results and above interpretation the following recommendations are proposed to Laxmi Concrete Products to enhance their inventory efficiency, cash flow stability and credit risk control.

4.2.1) Demand and Inventory Management Recommendations

1. **Prioritised Inventory control for Category A SKUs:** Since Category A SKUs contribute majorly to sales and revenue these SKUs should receive highest priority in procurement and stock monitoring. Stockouts for these SKUs would have cause excessive dip to revenue and sales.
2. **Controlled Stocking for Category C SKUs:** Since Category C SKUs like RCC Pipes and Bench and some Slabs contribute less then 5% to the revenue, maintaining their high inventory levels is not cost effective. Procurement must be done based on the demand to minimise the holding cost.
3. **Integration of Forecasting into planning:** Demand forecast results should be used as reference for monthly procurement of the SKUs. This can help to reduce excessive inventory and cost associated with it.
4. **Differentiated Safety Stock levels:** Category A SKUs should have higher safety stock level whereas safety stock level should be minimal for Category C SKUs. Uniform safety stock policy should be avoided.

4.2.2) Payment Default Risk and Receivables Management Recommendations

1. **Risk Based Client Segmentation:** Clients should be segmented based on the overall risk classification. Clients falling under No Risk and Low Risk categories may continue under existing credit terms, while Medium and High-Risk clients should be subjected to stricter monitoring.
2. **Revision of Credit terms for High-Risk clients:** Nearly two third of the clients are classified under High-Risk category so measures such as advanced payment, partial upfront collection should be considered to reduce the risk of payment delay or default. For example, 50% payment should be taken at the time of order, next 25% payment should be collected before the order is delivered and rest 25% payment should be collected at the time of delivery.
3. **Dedicated recovery focus on old dues:** Clients with old dues represents blocked capital. A recovery strategy should be implemented to address these legacy receivables. For example, clients with old dues must be contacted on weekly basis. Also, the firm should not accept new order from these clients until all the previous dues are cleared.
4. **Regular Monitoring of key indicators:** Client performance indicators like DSO, Collection Efficiency and Outstanding Amount proportion should be reviewed on a quarterly basis to early identify the risk of payment delays and defaults.
5. **Alignment of sales and Credit policies:** The firm should incorporate historical payment behaviour into credit approval and sales decision. This can help prevent accumulation of high risk receivables. For example, only clients falling under No risk and Low risk category should be allowed for credit sales.

Overall, this project shows how business data can help Laxmi Concrete Products take better decisions by analysing product demand, inventory contribution and customer payment behaviour. The study highlights key operational and financial challenges faced by the firm. The insights from this analysis support more organised inventory planning and improved cash flow at Laxmi Concrete Products.